A389474
Cubic reversion modulo nine and three

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Abstract

We construct the normalized integer solution of the defining equation, prove that its square is $x^2 + 2x^3$ modulo nine, and formalize the associated cubic-power assertion modulo three. The second assertion is a direct application of the classical Frobenius identity.

1 Source and conclusions

Let $U = x + O(x^2) \in \mathbb{Z}[[x]]$ be specified by

$$U(x^2 + 2x^3 + 2U(x)^2) = 3U(x)^2.$$

This is the A389474 source equation [1], with offset one. We first construct its unique integral solution. We then prove $U^2 \equiv x^2 + 2x^3 \pmod 9$ and, for every $k \geq 0$, $U^{3k} \equiv U(x^3)^k \pmod 3$. Powers without the composition symbol mean ordinary multiplication.

2 Integral construction and uniqueness

Theorem 2.1. *For every $s \in \mathbb{Z}$, there is a unique normalized series $R_s \in \mathbb{Z}[[x]]$ satisfying*

$$R_s(3x^2) = 2x^2 + R_s(x)^2 + 2sR_s(x)^3. \tag{1}$$

Proof. For any normalized integral series U, define its residual by

$$E_s(U) = U^2 + 2sU^3 - U(3x^2) + 2x^2.$$

Every coefficient of $E_s(U)$ is even. Indeed, after reduction modulo two, Frobenius gives $U(x)^2 = U(x^2)$ and $3x^2 = x^2$, so the residual vanishes. Also $E_s(U) = O(x^3)$ for every normalized U.

Suppose $V - U = cx^r + O(x^{r+1})$, with $r \geq 2$. The quadratic difference has leading term $2cx^{r+1}$; the cubic difference has order at least $r + 2$; and $(V - U)(3x^2)$ has order at least $2r \geq r + 2$. Consequently

$$E_s(V) - E_s(U) = 2cx^{r+1} + O(x^{r+2}). \tag{2}$$

Starting with $U = x$, choose its degree-r coefficient to cancel the degree-$(r+1)$ coefficient of the residual. The needed division by two is exact by the preceding parity argument. Equation (2) shows that earlier cancellations remain valid. Each degree is thus determined by finitely many earlier coefficients, and the resulting integral series has zero residual in every degree.

For uniqueness, take the first degree $r \geq 2$ where two candidate solutions differ. Equation (2) would give a nonzero degree-$(r+1)$ difference of residuals, because $2c \neq 0$ in $\mathbb{Z}$. This contradicts both residuals being zero. □

The proof formalizes divisibility of the residual by two before using integer division. Thus the construction does not assume the existence or integrality of an OEIS sequence. The names used below denote the uniquely constructed series.

3 The source bridge and the modulus-nine proof

Let $R = R_1$ and let U be its compositional inverse. Such an inverse exists over $\mathbb{Z}$ for every normalized series: its degree-n coefficient is determined with leading multiplier one, recursively. Both inverse identities are proved in the reused formal-series infrastructure. Substituting $U(x)$ in (1) gives

$$R(3U(x)^2) = 2U(x)^2 + x^2 + 2x^3.$$

Applying U proves the original A389474 equation

$$U\big(x^2 + 2x^3 + 2U(x)^2\big) = 3U(x)^2. \tag{3}$$

Conversely, if a normalized U satisfies (3), substitute its inverse R and then apply R to recover (1) for $s = 1$. Therefore Theorem 2.1 also gives existence and uniqueness for the literal A389474 source equation.

Theorem 3.1. *The A389474 series satisfies*

$$U(x)^2 \equiv x^2 + 2x^3 \pmod 9.$$

In particular, $[x^n]U(x)^2$ *is divisible by nine for every* $n > 3$.

Proof. Work in $(\mathbb{Z}/9\mathbb{Z})[[x]]$ and put $S = U^2$, $Q = x^2 + 2x^3$. The source equation becomes $U(Q + 2S) = 3S$. Let V be the genuine normalized compositional inverse of U over this ring. Apply V to obtain

$$Q + 2S = V(3S).$$

The series $3S$ has zero constant coefficient and is square-zero: $(3S)^2 = 9S^2 = 0$. A normalized series has the form $V(t) = t + t^2H(t)$, so $V(3S) = 3S$. It follows that $Q + 2S = 3S$, hence $S = Q$, as required. No division by three is used. □

This establishes a full formal-series identity, including the low-degree terms, and hence strengthens the source's statement restricted to degrees above three.

4 The standard Frobenius identity

For any integer formal series W and prime p, coefficient reduction gives

$$\overline{W}(x)^p = \overline{W}(x^p) \qquad \text{in } \mathbb{F}_p[[x]].$$

For each fixed coefficient one may truncate W at that degree. The polynomial identity $(u+v)^p = u^p + v^p$ in characteristic p, together with $a^p = a$ in $\mathbb{F}_p$, proves the displayed identity coefficient by coefficient. Thus it holds for every formal series, without a convergence assumption. This is standard algebra; its application below is not a new general principle.

Apply the identity at $p = 3$ to $W = U$ and raise both sides to the ordinary kth power. For every $n, k \geq 0$ this yields

$$[x^n](U(x)^{3k} - U(x^3)^k) \equiv 0 \pmod 3.$$

This proves the second selected source assertion, including the boundary cases $n = 0$ and $k = 0$. It requires no special equation for U.

5 Lean proof and exact scope

The dedicated entry file is `lean/Entries/A389474.lean`. Its declarations lie in namespace `FinalCollection.A389474`. The principal checked endpoints are:

- `source_equation`: the exact source equation.
- `source_existence_uniqueness`: existence and uniqueness over the integers.
- `integral_family_existence`: the entire integer-parameter auxiliary family.
- `square_mod_nine`: the full modulus-nine identity.
- `square_coefficients`: the source range $n > 3$.
- `power_coefficients`: the standard Frobenius consequence for all $n, k \geq 0$.

The full implementations are in `lean/More/CubicConsequences.lean`. The full integral construction, exact division by two, genuine compositional inversion, and both source-uniqueness directions are present in the shared implementation. No division by three is used modulo nine. Normalization and source uniqueness are explicit endpoints in the separate entry file.

The companion is a separate, explicitly typed Lean entry file. Its proofs invoke the preserved implementation modules; the complete dependency source is included in the same archive. Shared lemmas are reused rather than assumed. The entry file is not a self-contained replacement for Mathlib or the supplied common library.

The project pins Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The collection checker recompiles the proof sources and audits the theorems' transitive axiom dependencies. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. The delivered verification reports record the actual successful runs. Finite numerical checks are corroboration; the displayed conclusions are unbounded theorems. An AI review of source correspondence does not replace independent human review.

References

[1] OEIS Foundation, entry A389474, `https://oeis.org/A389474`. Source checked 24 October 2026. The source sequence and selected conjectures are credited to Paul D. Hanna.

[2] D. Goyal, supplied reference repository, `https://github.com/dvgyl/oeis-a396843-proof`. This separate presentation belongs to the final collected proof package; no earlier DOI is reassigned to it.